AROMATIC HYDROCARBON CONTAMINATION OF CLAY STRATA BELOW A PETROCHEMICAL SITE, UK

Organic Contaminant Migration in Clay Aquitards

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Abstract: Clay units have often been assumed to provide good protection for underlying aquifers from contamination. This assumption, however, is rarely proved. The capacity of clays to resist contaminant invasion in urban areas is of significant interest, especially where underlying aquifers are used for supply. We report the initial findings of an on-going field study to investigate organic contaminant penetration of clay strata below a former petrochemical site in the UK. The clay is a thin, relatively continuous unit 6-7.5m below ground surface within a shallow sandy aquifer contaminated with aromatic hydrocarbons (benzene, toluene, ethylbenzene, and styrene). Concentration/ depth profiles were obtained from cores at four locations and demonstrated transport of aromatic hydrocarbons into the clays with significant penetration in places. Preliminary modelling suggests that the clay penetration cannot always be explained by simple diffusion; advective transport through preferential pathways appears probable. Clay thin sections indicate the presence of palaeo-root holes and worm burrows. These potentially explain the enhanced solute penetration, and suggest that the presence of unconformities should be considered when assessing the protection afforded by thin clay beds.

Key words: clay; aquitard; aromatic hydrocarbons; solute transport; UK; preferential pathways

1. INTRODUCTION

Conservative dissolved solute movement in aquifers is controlled by advection and dispersion. Within low permeability deposits such as clays

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and mudstones, the advective component of groundwater flow may be small, and in such cases, depending on the geometry, diffusive transport can become important (*cf.* Johnson *et al.*, 1989). In aquifers, non-conservative solutes will also often be attenuated by sorption and degradation, and in particular by biodegradation. In many argillaceous units, sorption may be considerable because of the large proportion of clay minerals and organic matter often present: biodegradation in the clay matrix, however, may be limited by the inability of bacteria to enter the small pores.

Field and other evidence from various countries suggests that solutes can penetrate clays in some cases (e.g. Parker *et al.*, 2004). Fractures are common in clays, and may form preferential flowpaths. They may be produced during deposition, by tectonic movement, by shrinkage, or by freeze-thaw cycles. Millimetre- to centimetre-scale sand laminae ("stringers") may also provide higher permeability pathways. Contaminant migration through such features is particularly important for virus and bacterial transport, as size exclusion may prevent entry to the matrix. However, lateral diffusion into matrix blocks is often an important attenuation mechanism for solutes which do not have such size limitations, but in this case biodegradation may be limited within these blocks.

Non aqueous phase liquids (NAPLs) may migrate in the subsurface as separate immiscible phases. Given sufficient pressure head, dense NAPLs (DNAPLs) may migrate preferentially through the larger pore aperture pathways thereby penetrating clay horizons much more rapidly than dissolved-phase solutes, perhaps reducing residence time by orders of magnitude. Most hydrocarbons, however, are light non aqueous phase liquids (LNAPLs). Compared with DNAPLs, there is less likelihood of LNAPLs being in a situation to be driven through clay units below the water table, and often for LNAPL compounds solute penetration of clays is the main concern. This LNAPL conceptualization applies to the site described here.

Recent research on clays has focused on penetration of dense organic compounds in both NAPL and solute forms. Feenstra *et al.* (1991) calculated that 1-2 m of trichloroethene (TCE) penetration could occur into unfractured clay in a few decades by diffusion. Kueper and McWhorter (1991) and Parker *et al.* (2004; 1996) found much more rapid migration into clays via sand stringers and fractures. Fractures may only comprise a small volume in clays (0.001-1%) compared with the matrix porosity (30-70%), but form the main flow conduits (Freeze and Cherry, 1979). Rapid movement in fractures enables invasion to a much larger extent than in a non-fractured equivalent (Slough *et al.*, 1999). Relatively low DNAPL heads are required to enter small fractures, e.g. a head of 30 cm is necessary to enter fractures of 17μm aperture (e.g. O'Hara *et al.*, 2000).

This paper presents preliminary on-going research at a hydrocarbon-contaminated site. The aims are: (i) to examine the degree to which contaminant penetration has occurred in a shallow 1-2 m thick clay unit within a permeable sand aquifer; and (ii) to elucidate controlling processes such as the roles of diffusion, preferential advective pathways, and attenuation due to sorption and degradation. The site contains a shallow LNAPL source and the clay stratum studied some way below the water table has only been exposed to dissolved-phase concentrations. Thus migration of dissolved solutes, rather than NAPL, within the clay is the focus of the study.

2. STUDY SITE

2.1 Setting

The precise location of the site cannot be disclosed. However, it is in the UK, and was formerly occupied by a large petrochemical works that is currently undergoing phased remediation. Topographically, the site is flat lying, occupying coastal plains approximately 1 km from the sea at 6-10 m AOD [above Ordnance Datum (~sea level)]. Shallow geological deposits in the study area comprise 0.2 m of made ground underlain by about 6 m of fine to medium windblown sand with occasional peat horizons. Grey clay of 0.5-2 m thickness underlies these deposits. This clay is composed of illite and chlorite with some visually identifiable organic-rich zones. Its exact lateral extent and continuity is unknown, but appears laterally continuous over several hundred metres at least. The geological history suggests that the clay is of lacustrine origin. Discontinuous thin clay lenses (up to 5-10 cm thick) were found above and below this clay bed. The deeper aquifer consists of fine to medium grained marine sand with occasional lenses of silt and clay. It occurs down to about 25 m bgs (below ground surface), below which till and upper Carboniferous mudstones occur.

The site has a shallow water table at 0.7-2m. Groundwater flow is radial away from an apparent recharge mound, central to the study area. Groundwater therefore flows locally away from the site in all directions, though regionally flow is towards the coast. The hydraulic conductivity of the sands is approximately 35 m/d. Site investigation by consultants (Celtic Technologies) has indicated that the head in the overlying sands is typically around 1m higher than that in the sands below the clay bed at ~6 m bgs, suggesting that the clay is a barrier to vertical flow and is relatively continuous. The hydraulic gradient across the clay is around 0.4 (downwards).

2.2 Extent of Contamination

Prior to study, it was known that the site had been subject to contaminant releases from the former hydrocarbon storage tanks and production plant areas. Site investigations by consultants have focused upon the shallow sand aquifer. LNAPL oils floating on the water table were detected in a number of boreholes with residual LNAPL contamination in the first 3 m bgs. More limited water quality data from occasional wells screened below the ~ 6 m bgs clay layer confirmed some low-concentration dissolved-phase contamination by aromatic hydrocarbons in the deeper sand aquifer. The research study described herein focuses upon the processes controlling the transport of the aromatic hydrocarbon solutes across this 1-2 m thick clay horizon to the deeper sand aquifer unit.

3. METHODOLOGY

3.1 Sampling

Cores of 90 mm diameter were collected from four locations using a sonic bore drilling rig (Drill Corp, County Durham, UK). The cores were recovered in plastic sleeves, which were cut open longitudinally immediately before sampling in the field. Soil samples for water extraction were taken every 10 cm and placed in 40 ml vials. Vials were completely filled with aquifer material from the centre of the core and stored in cool boxes before being sent away for commercial analysis. Soil samples for methanol extraction were taken every 5cm using a stainless steel 1cm ID sub-corer. The 5cm long sub-cores were taken through the centre of the core and were placed in 15ml of pre-weighed methanol in 40 ml vials on site. Vials were stored at 4°C in the laboratory before analysis. Additional core material was wrapped in plastic in cool boxes, then stored at 4°C in the laboratory.

3.2 Soil Core Analysis Method

Contaminant concentrations in soil samples from cores 3 and 4 were analyzed using water-extraction analysis whereas cores 1 and 2 were analyzed using methanol extraction. The former quantifies the dissolved aromatic hydrocarbon contamination and probably some of the initially sorbed phase contamination, whereas the latter is a more aggressive extraction methodology and measures the total aromatic hydrocarbon contamination present in the soil sample.

In the case of the methanol extraction method, samples of approximately 10 g (exact mass later determined) were collected at 5-10 cm intervals from the core and placed in 15 ml of pre-weighed methanol with a surrogate compound in 40 ml vials. The sub-corer was washed with Decon®, organic-free water and methanol between samples. Samples were stored at 4˚C for at least two weeks prior to analysis and were daily shaken mechanically to enhance clay dispersion. Before analysis, samples were shaken and then centrifuged at 2500 rpm for 4 minutes. A 100 µl aliquot was taken and added to 15ml of organic-free water in a 20 ml headspace vial. Samples were analyzed by GC-MS in both the regular scanning mode and also selective ion monitoring mode depending upon the concentrations detected.

Analysis of extracts was by headspace-GCMS. Analysis was performed using an Agilent Model 6890 series gas chromatograph (GC) equipped with a Gerstel MPS2 autosampler and a HP 5973 inert mass spectrometer (MS). Chromatographic data were collected and handled by Chemstation software. The headspace autosampler conditions were set as follows: 2.5ml HS syringe; sample volume, 500µl; incubation temperature, 60˚C; incubation time, 15mins; agitation speed, 750rpm. A 30m x 0.32mm ID DB624 column with 1.80µm film thickness (J and W Scientific) was utilized. The carrier gas was helium at a constant flow rate of 3ml/min. The injector was held at 50˚C and a split ratio of 10:1 was used. The column oven temperature programme began at 60˚C for 0.5 minutes, increased at 35˚C/min until a final temperature of 230˚C. The mass spectrometer transfer line was set at 280˚C.

Standard concentrations of benzene, toluene, ethylbenzene and styrene were made by dilution with methanol from pure-phase chemicals. Headspace standard solutions were prepared by adding 100 µl of the working standard into a screw-capped vial already amended with a volume of organic-free water.

Water extraction samples were analyzed by a commercial laboratory (Severn Trent Laboratories (STL)). Samples were collected in completely-filled 40 ml vials. The analysis method is based on the US EPA Methodology 5021 (USEPA, 2004a). Samples of 2.5 g were mixed with 8 ml of deionised water, sodium chloride and sand, aiding the partitioning of VOCs into the gas phase. Samples were also analysed using headspace GC-MS.

3.3 Partitioning Calculations for Water Concentration

Analysis of the soil samples yields a mass of organic contaminant per unit mass of soil. Concentrations predicted to be present in an equilibrated pore water in the soil sample may be back-calculated from the soil sample analysis results via the use of a partitioning equation (Eq. 1) (e.g. Feenstra *et al*., 1996):

$$C_w = \frac{C_t \rho_b}{\left(K_d \rho_b + \phi_w + H_c \phi_a\right)} \tag{1}$$

where C_w is the concentration in the porewater (mg/l), C_t is the total concentration from the solid sample analyses (µg/g dry weight), ρ_b is the dry bulk density (g/cm^3), K_d is the partitioning coefficient (ml/g), ϕ_w is water filled porosity (volume fraction), H_c is the dimensionless Henry's Law Constant, and ϕ_a is the air filled porosity (volume fraction). The measured, estimated, or calculated values for the above parameters are given in Table 1.

Table 1. Estimated parameters used in the calculation of porewater concentrations [[1] USEPA (2004b); [2] Fetter (1999); [3] Parker et al. (2004); [4] calculated; [5] calculated from field data; [6] Morris and Johnson (1967)]

Chemical	H_c [1]	K_{oc} (ml/g) [2]	f_{oc} [3]	K_d (ml/g) [4]	ρ_b (g/ml) [5]	ϕ_w [6]
Benzene	2.28×10^{-1}	66	0.001	0.066	1.62	0.3
Toluene	2.72×10^{-1}	145	0.001	0.145	1.62	0.3
Ethylbenzene	3.23×10^{-1}	207	0.001	0.207	1.62	0.3
Styrene	1.13×10^{-1}	912	0.001	0.912	1.62	0.3

4. RESULTS AND DISCUSSION

4.1 Shallow Aquifer Profiles

Geological log descriptions from the four cores are given in Table 2. Cores 1, 2 and 3 were taken within an area of 20m^2, whilst core 4 was taken from approximately 165 m away. It is unknown whether the main thick clay beds, appearing between 2.39 and 3.95 m AOD, to a thickness of between 0.35 and 0.65 m, are continuous. In cores 1 and 2, additional thin clay lenses occur around the main clay bed.

Table 2. Geological log descriptions for cores taken.

Core	Depth (m bgs)	Log
1	9.85-3.95	Upper profile, fine SAND
	3.95-3.56	Grey clay
	3.56-3.30	Fine sands
	3.3-3.25	Thin, grey clay lens
	3.25-3.00	Fine sands
2	9.84-3.39	Upper profile, fine SAND
	3.39-3.34	Dense, fine sands with lens of clay
	3.34-3.04	Fine sands
	3.04-2.39	Dark grey clay
	2.39-2.34	Fine sands
3	9.65-3.655	Upper profile, medium SAND
	3.655-3.155	Brown-grey, soft to firm, mottled CLAY
	3.155-2.655	Dark brown, coarse-medium, loose, silty SAND
4	9.584-3.684	Upper profile, medium SAND
	3.684-2.584	Dark grey, firm CLAY. Rare black organics. Thinly laminated.

The highest concentrations of benzene were found at the water table which is known to reside between 1 and 2m (Figure 1). The maximum calculated water concentration of benzene in core 3 at this level was 963 mg/l, which was below the solubility limit of 1750 mg/l, and hence it is assumed to be in dissolved form. Bailer samples showed free-phase LNAPL to be floating on the water table; however this was thought to be comprised of ethylbenzene, styrene, and toluene primarily.

4.2 Clay Profiles

Detailed benzene profiles in the clay bed for the four cores are shown in Figure 2. Benzene penetration occurred into all of the clay units to some degree, with maximum penetration occurring in the clay of cores 1, 2, and 3. Cores 3 and 4 have significantly lower contaminant levels than cores 1 and 2; this is ascribed to the different extraction method used (water for cores 3 and 4, methanol for cores 1 and 2). The partitioning calculations do not suggest the presence of free-phase LNAPL since the greatest equivalent porewater concentrations were calculated to be ~ 40 mg/l (in core 2), significantly below benzene solubility. Cores 1 and 3 show that contaminant mass was highest near the sand-clay interface and decreased with depth from this point.

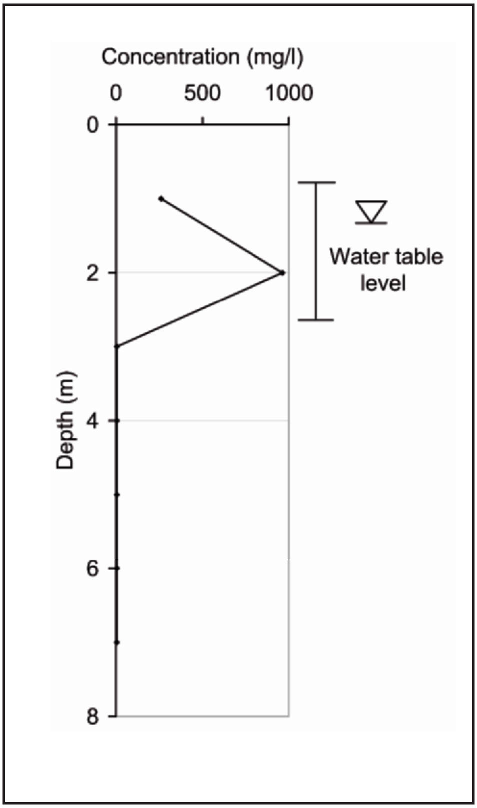

Figure 1. Benzene concentrations in the porewater of core 1 with depth (porewater concentrations based on partitioning calculations on soil concentration data).

Core 1 (Figure 2) shows a profile that is qualitatively consistent with that expected from diffusion-dominated transport, although the clay stratum is thin at this location. Core 2, in contrast, shows higher concentrations extending throughout a much thicker portion of the clay suggesting that advection may be important. Thin clay lenses were also identified in cores 1 (2 cm thick) and 2 (8 cm thick) around the main clay stratum. Benzene concentrations above these clay lenses were found to be higher than those below, suggesting that they may be partial barriers to contaminant migration.

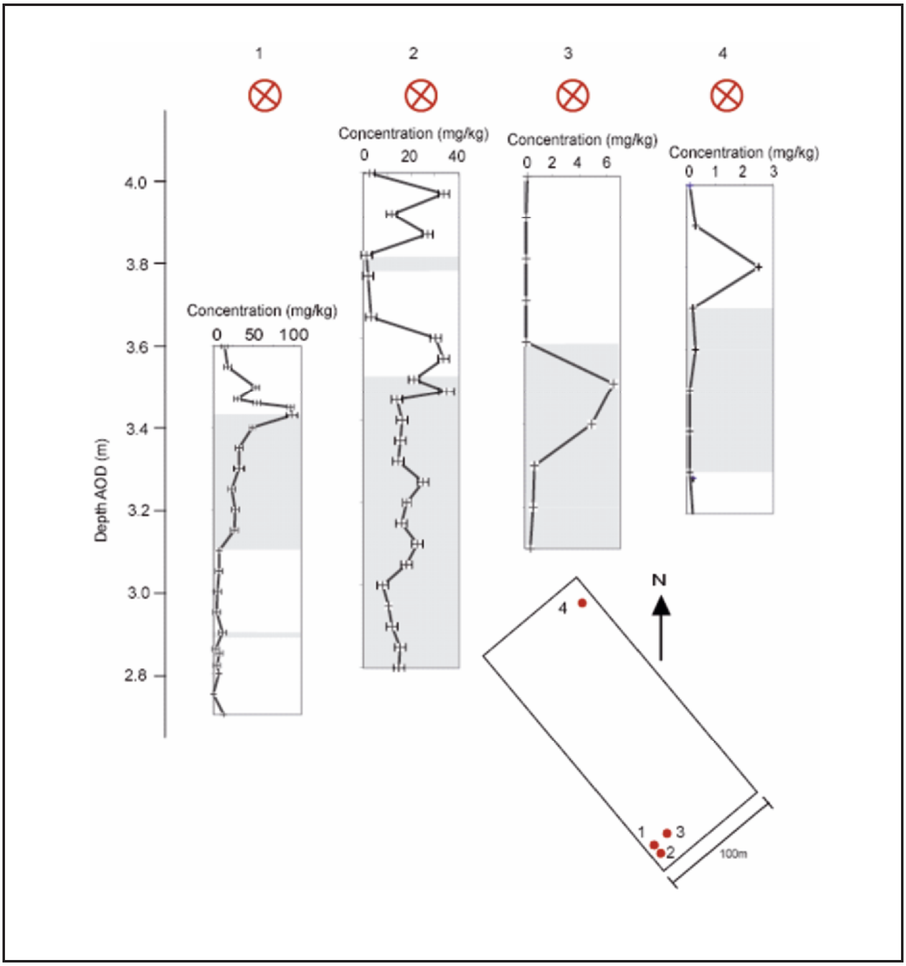

Figure 2. Benzene depth soil-concentration profiles for cores 1-4 (grey bands denote the clay strata).

5. 1-D CONTAMINANT TRANSPORT MODELLING

The movement of non-degrading organic contaminants into the non-fractured clay beds may be expected to be controlled by diffusion and sorption. In more fractured, preferential-pathway clays, advection and dispersion processes would assume greater significance (Parker *et al.*, 2004). A simple 1-D model was used to assess processes controlling contaminant migration into the main clay stratum in core 1 for which the contaminant distributions appeared to be most diffusion-like. Simulated and core data are shown in Figure 3 for benzene.

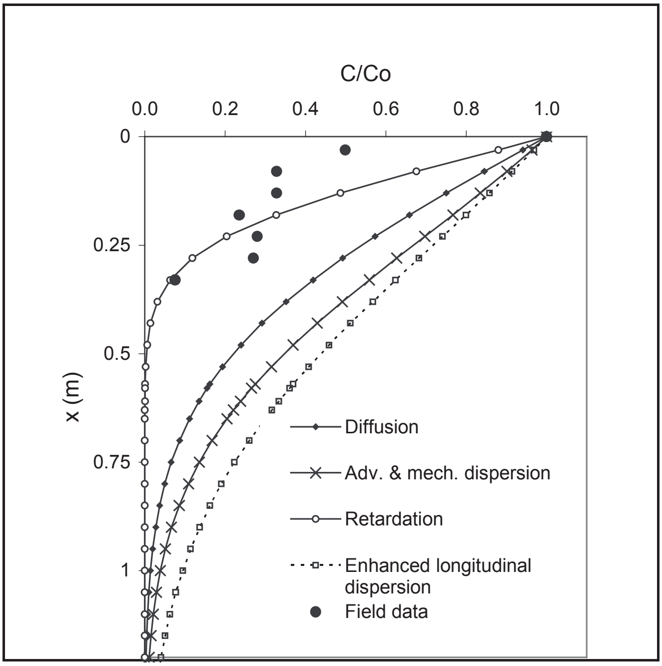

Figure 3. Normalized dissolved-phase concentration profiles for benzene core 1 data and simulated: diffusion-only profile (t = 8.2y, D* = $2.8x10^{-5}$ m^2/d); advection + dispersion profile (K = $7.1x10^{-6}$m/d, i = 2.8, n = 0.4, v = $5.1x10^{-5}$ m/d, t = 8.2y, α_L = 0.035m, D* = $2.8x10^{-5}$ m^2/d); advection with retardation and large dispersivity (retardation factor = 6 based on laboratory sorption experiments; α_L = 0.35m).

Figure 3 shows contaminant profiles produced by simulating the following conditions: diffusion only; advection + dispersion; and advection + dispersion + retardation. A further profile was calculated using the dispersion flow solution, but with an increased longitudinal dispersion value to represent in a crude way the presence of preferential pathways. The profiles were calculated assuming 1-D movement into an infinite medium from a constant concentration source, where: for $x = 0$, $t \geq 0$, $C = C_o$; for $0 < x < \infty$, $t = 0$, $C = 0$; $0 < x < \infty$, $t > 0$, $C = C$; and for $x = \infty$, $t \geq 0$, $C = 0$. The time period was estimated from site records; average linear velocity was estimated using Darcy's Law with an hydraulic conductivity (K) of $2.6x10^{-3}$ m/y, a gradient (i) of 2.857, and a porosity (n) of 0.4; longitudinal dispersivity (α_L) was estimated at 0.035m; and the diffusion coefficient (D*) was estimated as $2.77x10^{-5}m^2/d$.

As can be seen from Figure 3, the penetration of benzene into the clay of core 1 can be described by a retarded diffusion profile. The degree to which advection would be expected to occur in the clay matrix and influence penetration is shown to be relatively small, and cannot explain profiles such as that seen in core 2.

6. PREFERENTIAL PATHWAYS IN THE CLAY

Further work is underway to identify possible heterogeneities within the clay which may explain the core 2 where contaminant penetration is greater than can be explained by simple diffusion or matrix advection. Studies of pesticide movement through preferential flow paths by Jørgensen *et al*. (2002) found 96-98% of flow to be along fractures and Broholm *et al.* (1999) showed that the transport of dissolved phase low molecular weight organic compounds through fractured clayey till may occur as rapidly as bromide, a conservative tracer.

Horizontally-oriented thin sections of the clay allowed the identification of possible preferential flowpaths as shown in Figures 4 and 5. These figures include examples of voids surrounded by a layer of black material of possible root wall origin. Also, there is evidence of mineral alteration, both suggesting the presence of root channels formed prior to the deposition of the upper sand. Such features may cause contaminant behaviour changes in terms of both flow velocity and sorption. Organic-rich root-hole linings may increase sorptive retardation of contaminants, much as Turner and Steele (1988) reported for metals in organic-lined fractures. Weathering reactions may also change the characteristics of the clay surrounding the root-holes, for example in terms of the surface area available for sorption.

Evidence of worm activity, in the form of cloud-shaped excreta/bioturbation can also be seen in Figure 5. The properties of such material may vary from that of the bulk clay in respect of sorption, flow, and biodegradation potential. Similarly, intact pieces of organic matter, as also seen in Figure 5, may cause local variation in organic contaminant transport behaviour. The distribution of these features with depth has yet to be established, but it would be expected that they would be more common in the upper parts of the clay. It is also expected that the thinner the clay bed, the greater the chance that the features will fully penetrate.

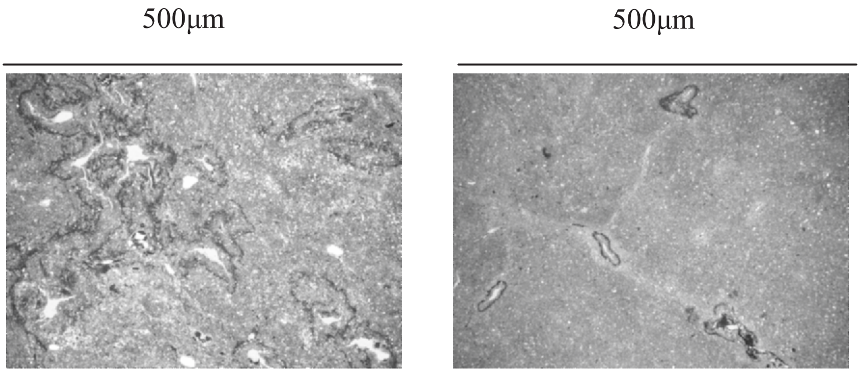

Figure 4. White voids (left) and mineral alteration (right) indicating the presence of root holes.

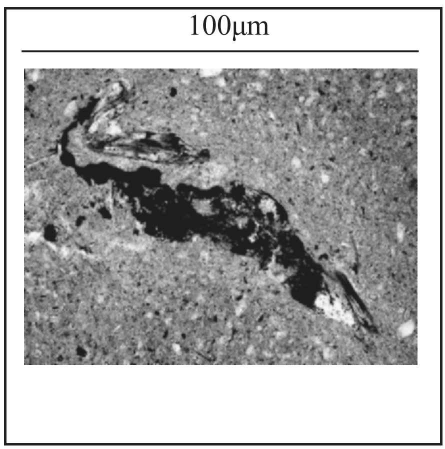

Figure 5. Black cloud-shaped structure indicating worm activity.

7. CONCLUSIONS

The penetration of dissolved-phase aromatic hydrocarbon solutes from a sandy aquifer into a 0.3-1.5 m thick bed of lacustrine clay at 6 m depth was investigated using core-based contaminant profiling. Penetration into the clays occurred to over 0.6 m and could not be fully explained simply by 1-D diffusion or advection-dispersion (-retardation) modelling over the anticipated contaminant contact with the clay.

The contaminant profiles were variable over a small area. Of the two most detailed profiles obtained, one had a profile consistent with diffusion-only transport. The other profile cannot be explained by diffusion alone and it is believed heterogeneities within the clay layer creating preferential flowpaths have influenced contaminant transport. Observations of palaeo-roots in the clay offer a likely explanation. Further field cores and other relevant data (e.g. further hydraulic head gradient analysis over the clay layer) as well as supporting laboratory studies (e.g. sorption) are continuing. It is concluded that unconformity surfaces should be taken into account when deciding whether a clay bed affords protection for an underlying aquifer.

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